## Aim:To Implement Linear Regression

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
df = pd.read_csv("HousingData.csv")
```

```
df.head
```

**pandas.core.generic.NDFrame.head**
def head(n: int=5) -> Self

```
3       lion
4     monkey
5     parrot
6      shark
7      whale
8      zebra
```

```
df.tail
```

**pandas.core.generic.NDFrame.tail**
def tail(n: int=5) -> Self

```
3    lion
4  monkey
5  parrot
6   shark
7   whale
8   zebra
```

```
df.isnull
```

**pandas.core.frame.DataFrame.isnull**
def isnull() -> DataFrame

```
>>> ser.isna()
0    False
1    False
2     True
dtype: bool
```

```
df.info
```

**pandas.core.frame.DataFrame.info**
def info(verbose: bool | None=None, buf: WriteBuffer[str] | None=None, max_cols: int | None=None, memory_usage: bool | str | None=None, show_counts: bool | None=None) -> None

```
---   ------    --------------    -----
 0   column_1  1000000 non-null  object
 1   column_2  1000000 non-null  object
 2   column_3  1000000 non-null  object
dtypes: object(3)
memory usage: 165.9 MB
```

```
df.shape
```

```
(506, 14)
```

```
X = df[['RM']] # Input (Number of rooms)
y = df['MEDV'] # Output (House price)
```

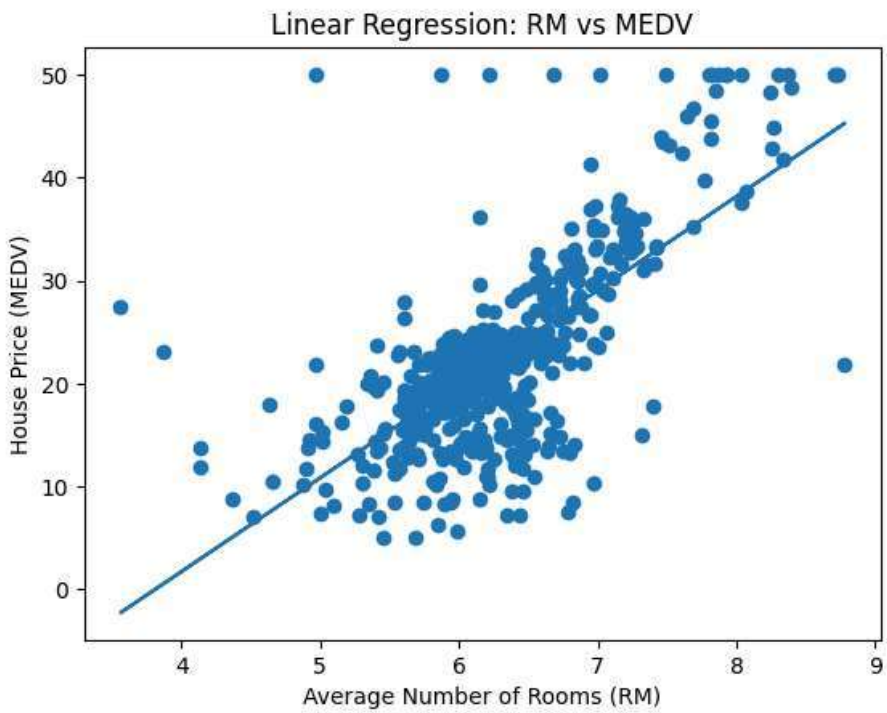
```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred = model.predict(X)
```

```
plt.figure()
plt.scatter(X, y) # Actual data points
plt.plot(X, y_pred) # Regression line
plt.xlabel("Average Number of Rooms (RM)")
plt.ylabel("House Price (MEDV)")
plt.title("Linear Regression: RM vs MEDV")
plt.show()
```




```
print("Slope:", model.coef_)
```

```
Slope: [9.10210898]
```

```
print("Intercept:", model.intercept_)
```

```
Intercept: -34.67062077643857
```